📊 Excel Data Analyst Project Documentation – E-Commerce Sales Analysis

Project Overview

This project focuses on analyzing an E-Commerce Sales dataset using Microsoft Excel. The objective was to clean, organize, analyze, and visualize sales data to uncover meaningful business insights. The project demonstrates practical Excel skills in data cleaning, formula usage, Pivot Tables, dashboards, and business reporting.

Business Problem

The business needed to understand sales performance across different categories, brands, customer demographics, and inventory levels. The analysis aimed to identify high-performing products, customer purchasing patterns, and inventory distribution to support better business decisions.

Dataset Information

Dataset: E-Commerce Sales Dataset

The dataset includes the following information:

- Order ID
- Customer ID
- Product Category
- Brand
- Product Name
- Gender
- Age
- Loyalty Level
- Sales Amount
- Cost
- Stock Quantity
- Order Date
- Region

Data Cleaning Process

The dataset was cleaned and prepared before analysis using Excel.

Data cleaning steps included:

- Removed duplicate records.
- Checked and handled missing values.
- Corrected inconsistent text formatting.
- Standardized category and brand names.
- Converted data into appropriate formats (Number, Currency, Date).
- Verified data accuracy and consistency.
- Created an Excel Table for efficient analysis.

Excel Functions Used

The following Excel functions were used during the analysis:

- SUM()
- AVERAGE()
- COUNT()
- MIN()
- MAX()
- IF()
- SUMIF()
- COUNTIF()
- AVERAGEIF()
- RIGHT()
- LEFT()
- MID()
- UPPER()
- LOWER()
- TRIM()
- CONCAT()
- TEXT()

Pivot Table Analysis

Several Pivot Tables were created to summarize business performance, including:

- Category-wise Cost
- Category-wise Stock
- Brand-wise Cost
- Gender vs Loyalty Level
- Age vs Loyalty Level

These Pivot Tables enabled quick aggregation and comparison of key business metrics.

Visualizations Created

The dashboard includes multiple visualizations such as:

- Column Charts
- Bar Charts
- Pie Charts
- Slicers
- KPI Summary Cards

These charts help users quickly understand trends and compare performance across different business dimensions.

Dashboard Components

The interactive dashboard contains:

- Total Sales KPI
- Total Cost KPI
- Total Stock KPI
- Category-wise Sales & Cost
- Brand Performance
- Customer Gender Distribution
- Loyalty Level Analysis
- Inventory Distribution
- Interactive Slicers for filtering data

Key Business Insights

- Electronics recorded the highest total cost, while Beauty had the lowest.
- Sports maintained the highest stock inventory among all categories.
- Home category had the lowest available stock.
- Hill PLC showed the highest brand-wise cost, while Davis PLC recorded the lowest.
- Female customers had a slightly higher loyalty count than male customers.
- Customers aged 46 showed the highest loyalty level among the selected age groups.
- Inventory and spending were distributed relatively evenly across categories, indicating balanced business performance.

Recommendations

- Optimize inventory planning by maintaining stock levels for high-demand categories like Sports and Electronics.

- Review inventory strategy for lower-stock categories to prevent shortages.
- Strengthen customer loyalty programs targeting both male and female customers.
- Continue monitoring high-cost brands to maximize profitability.
- Use customer demographic insights to create targeted marketing campaigns.
- Regularly update dashboards for real-time business monitoring and decision-making.

Skills Demonstrated

- Microsoft Excel
- Data Cleaning
- Data Validation
- Excel Tables
- Pivot Tables
- Pivot Charts
- Dashboard Design
- Data Visualization
- Business Analysis
- Data Interpretation
- Reporting
- Analytical Thinking

Conclusion

This project demonstrates how Microsoft Excel can be used to transform raw E-Commerce sales data into meaningful business insights. By applying data cleaning techniques, Excel functions, Pivot Tables, and interactive dashboards, the analysis supports informed decision-making, improves reporting efficiency, and highlights opportunities to optimize sales performance, inventory management, and customer engagement.